Merge Binary Trees

AlgoExpert · Medium · Binary Trees

The problem

- Overlay two trees; overlapping nodes sum their values.
- Mutates and returns tree1.

Intuition

- Overlay the trees node by node: add values where both exist, otherwise keep whichever exists.

Approach

- 1. If one tree is null, return the other
- 2. Add the values
- 3. Merge left children and right children

Java solution

```java
public class MergeBinaryTrees {
    public static BinaryTree mergeBinaryTrees(BinaryTree a, BinaryTree b) {
        if (a == null) return b;
        if (b == null) return a;
        a.value += b.value;
        a.left = mergeBinaryTrees(a.left, b.left);
        a.right = mergeBinaryTrees(a.right, b.right);
        return a;
    }
}
```

Complexity

- Time: O(n) of the smaller overlap
- Space: O(h)

Recap

- Overlay the trees node by node: add values where both exist, otherwise keep whichever exists.
- Complexity: O(n) of the smaller overlap